

PATIENT MULTI-OMICS GENOMIC HEALTH SUMMARY

Patient Submitter ID: TCGA-B0-4690 | Tumor: Kidney Renal Clear Cell Carcinoma (KIRC) | Stage: Stage III

TCGA Project:	TCGA-KIRC	cBioPortal Study:	kirc_tcga_pan_can_atlas_2018
Gender / Age:	Male / 70 yrs	Vital Status / OS:	Alive (21.5 mo)
Hybrid ML Risk:	81.2% (CRITICAL RISK)	VQC / Classical:	78.0% / 83.4%

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

Gene	Locus (GRCh38)	Transcript	Protein Change	ClinVar Status	COSMIC ID	gnomAD Pop AF	VAF %
PTEN	chr10:10q23.31	ENST00000371953	p.R130G Missense Mutation	Pathogenic VCV000014382	COSV51719875 FATHMM: 0.99 Pathogenic	0.000002	41.3%

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ_{VQC} / R_Y / CNOT)

Gene Symbol	Canonical Ensembl ID	VAF Hotspot Frequency	Variational Angle θ (Radians)	Qubit Hilbert Space Projection
PTEN	ENSG00000171862	41.3%	$\theta = 2.1150 \text{ rad (121.2}^\circ\text{)}$	$ \psi\rangle = \cos(1.058) 0\rangle + \sin(1.058) 1\rangle$

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:

VHL tumor suppressor pathway surveillance, pembrolizumab + axitinib dual anti-angiogenic/checkpoint regimen, and renal retroperitoneal imaging.

Genomic Biomarker Summary: Identified somatic hotspot mutations in PTEN (p.R130G). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the Blazeфинix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.